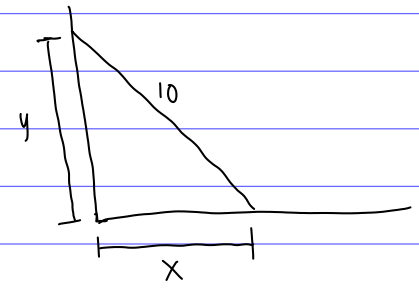
6



$x = x(t)$

$y = y(t)$

$x^2 + y^2 = 10^2$ $\quad \frac{dx}{dt} = 1.$

(a) Find $\frac{dy}{dt}$ when $x=6$

$x^2+y^2 = 100$

$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$

Side $6^2+y^2=10^2$

$y^2 = 100-36 = 64$

$y = 8$

$2(6)\cdot 1 + 2(8)\cdot \frac{dy}{dt} = 0$

$16 \cdot \frac{dy}{dt} = -12$

$\frac{dy}{dt} = -\frac{12}{16} = -\frac{3}{4}$ ft/sec (Why negative?)

When the bottom is 6 ft from wall, the top slides down at a rate of $\frac{3}{4}$ ft/sec.

(b) Find $\frac{dA}{dt}$ when $x=6$ $\quad A = \frac{1}{2}xy$

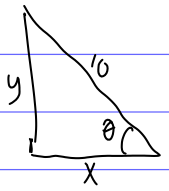
$\frac{dA}{dt} = \frac{1}{2}\cdot\left(x\cdot\frac{dy}{dt} + y\cdot\frac{dx}{dt}\right)$

$16-9=7$

$= \frac{1}{2}\left(6\cdot -\frac{3}{4} + 8\cdot 1\right) = \frac{1}{2}\left(-\frac{9}{2}+8\right) = \frac{1}{2}\left(\frac{7}{2}\right) = \frac{7}{4}$

When the bottom is 6 ft from wall, the area increases at a rate of $\frac{7}{4}$ ft^2/sec.

(c) Find $\frac{d\theta}{dt}$ when $x=6$



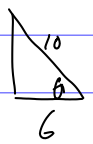
$\cos(\theta) = \frac{x}{10}$

$10\cos\theta = x$

$\downarrow \frac{d}{dt}$

$10\cdot -\sin\theta \cdot \frac{d\theta}{dt} = \frac{dx}{dt}$

Side:



$\cos(\theta) = \frac{6}{10}$

$\theta = \cos^{-1}\left(\frac{6}{10}\right)$

$10\cdot -\sin\left(\cos^{-1}\left(\frac{6}{10}\right)\right)\cdot\frac{d\theta}{dt} = 1$

$\frac{d\theta}{dt} = \frac{1}{-10\sin\left(\cos^{-1}\left(\frac{6}{10}\right)\right)} \approx -0.125$

When the bottom is 6 ft from wall, the angle decreases at the rate of 0.125 radians/sec.